An Open-Source Tool for Lifetime Estimation of IGBT–Diode Power Modules: A Module in PEARL

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***Abstract*—This work presents an IGBT-diode power module reliability assessment framework developed within PEARL (Power Electronics Age and Reliability), an open-source Python framework for the lifetime assessment of power electronics in inverter-based resources. The framework establishes a unified computational pipeline that links time-varying mission profiles, operating conditions, design configurations and component specifications to semiconductor lifetime by integrating electrical loss modelling (switching and conduction losses), dynamic thermal simulation (multi-stage Cauer-type RC networks) and lifetime estimation (rainflow-based thermal cycle extraction combined with physics of failure lifetime models and Miner's damage accumulation). The implementation is modular and computationally efficient, enabling long-duration simulations with direct datasheet-driven parameterization of IGBT-diode power modules, thermal interfaces and heat sinks. Case studies show that reactive power operation significantly alters device-level lifetime, with the module lifetime being approximately one year at both pf=1 and pf=0, while a maximum lifetime of about 19 years is achieved around power factor of 0.4 due to balanced thermal loading of the IGBT and diode. Furthermore, the framework is applied to evaluate heat sink configurations, demonstrating that thermal design directly influences junction temperature and semiconductor lifetime, with the 10U configuration at 4 m/s maintaining the lowest junction temperature and achieving the longest lifetime of approximately 4 years. The results highlight the importance of reliability-aware operation and design in converter-dominated energy systems and position PEARL as a transparent and scalable tool for early-stage reliability assessment.**

***Index Terms*—Power-electronics reliability, IGBT-diode module, electro-thermal modelling, mission profile, lifetime estimation**

I. Introduction

The ongoing transition toward converter-dominated power systems has made the reliability of power electronic converters central to grid resilience as they increasingly form the interface between generation, storage and the grid. In solar PV plants, inverters account for over 60% of system failures, corresponding to an estimated \$2.5 billion in annual losses [1], and in wind turbines, power electronic converters remain a primary source of downtime [2]. At the same time, inverter-based resources (IBRs) are no longer passive grid followers but are increasingly required to provide grid critical services such as voltage regulation, reactive power support and fault ride through [3]. These services impose additional electrical and thermal stresses on semiconductor devices within converters, accelerating aging and amplifying reliability concerns [4], [5], [6]. Collectively, these observations highlight power electronics as a key reliability bottleneck in the energy transition, underpinning essential grid support functions and challenging long-term grid planning and asset economics [1], [7]. As the deployment and market participation of IBRs continue to expand [8], robust reliability assessment of power electronic components becomes essential for long-term asset performance and system-level planning.

At the heart of these reliability challenges lie the IGBT-diode power modules that form the switching core of modern converters. The dominant failure mechanisms in IGBT-diode power modules arise from deterministic wear out processes driven primarily by thermo-mechanical fatigue, which is induced by electrical loading that causes repetitive junction temperature cycling [9]. Field data indicate that temperature induced stress accounts for more than 55% of failures in power-electronic systems [10]. Such failures are induced by repetitive junction temperature cycling caused by fluctuating electrical loading and environmental conditions, which progressively degrade bond wires, solder layers and interconnect structures over time [9].

A. Identified Research Gap

The growing penetration of power electronic converters exposes a key tooling gap, namely that there is still no unified, transparent and reproducible framework that maps realistic, time-varying mission profiles to quantitative component lifetime consumption. As summarized in Table I, existing reliability approaches are fragmented. For example, high-fidelity methods (e.g., FEM and accelerated aging tests) provide detailed insight, but are too computationally expensive for long-duration, system-level mission profiles [11], while manufacturer tools are convenient and device calibrated but closed-source, limited in scope and difficult to integrate into broader operational studies. Consequently, despite the central role of power electronics in providing grid critical functions, an accessible mission profile driven reliability framework remains missing.

B. Paper Contribution

This paper presents an open-source tool called PEARL, an electro-thermal reliability framework for power electronic systems, with this work specifically focused on IGBT-diode power modules. In contrast to existing tools that are computationally

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TABLE I: Comparison of available commercial and open-source tools for power electronics lifetime / RUL / electro thermal analysis.

Tool	Type	Main capabilities	Strengths	Weaknesses / watchouts
CeRULEo	Open-source / Python	Train/evaluate RUL ML models	Good for ML benchmarking	No device to mission physics chain
ProgPy / NASA libs	Open-source / Python	PHM / prognostics components	Useful for RUL algorithm validation	Not power electronics specific
Infineon IPOSIM	Manufacturer-proprietary	Device-specific lifetime estimation	Easy, vendor calibrated	Closed-source; limited to Infineon parts
Semikron / SemiSel	Manufacturer-proprietary	Loss/thermal calculators; lifetime curves	Manufacturer recommended curves	Lacks system-level RUL integration
PLECS (Plexim)	Commercial (simulation)	Electrical-thermal co-simulation	Well integrated; widely used	Requires custom RUL integration
ANSYS (Icepak / Sherlock)	Commercial (FEM)	3D CFD/thermal + physics of failure	High-fidelity FEM, thermomechanics	Heavy, costly; not real-time
COMSOL	Commercial (FEM)	3D multiphysics thermo-mechanical	Excellent for hotspot/stress validation	Slow; not for real time RUL
MathWorks RUL Toolbox	Commercial (toolbox)	Data driven, Simulink integration	Rich ML + model hybrid tools	License cost; needs physics integration
PEARL	Open-source / Python	Unified electrical, loss, thermal and lifetime modelling; datasheet-driven component parameterization	Reproducible, scalable; flexible across operating conditions and design configurations	Depends on empirical lifetime models; not a FEM replacement

intensive, device-specific or closed-source, PEARL provides a unified and reproducible workflow that links time-varying mission profiles, operating conditions, design configurations and component specifications to semiconductor lifetime estimation by integrating electrical loss modelling, dynamic junction temperature evolution and lifetime estimation as highlighted in Table I. The framework is computationally efficient, transparent and scalable, enabling reproducible research and reliability-aware design.

C. Paper Organization

Section II presents the proposed methodology, Section III describes the case studies used for evaluation and Section IV discusses the corresponding results and key findings. Finally, Section V concludes the paper, while Section VI outlines directions for future work.

II. Methodology

This section describes the methodology of the IGBT–diode power module reliability assessment framework developed within PEARL. The framework combines electrical loss, thermal and lifetime models into a single reproducible lifetime assessment pipeline, which is described step by step in the remainder of this section.

A. Electrical Loss Modelling

IGBT and diode power losses are computed from the mission profile operating points and the inverter configuration by first reconstructing the instantaneous device current waveforms through the IGBT and its anti-parallel diode. Based on these currents, conduction and switching losses are evaluated using the established formulations in [4], yielding time series power loss profiles for the IGBT and diode across the mission profile. All required loss model parameters are extracted directly from the manufacturer datasheets of the selected semiconductor module. For simplicity and reduced computational effort, the parameters taken from manufacturer datasheets are assumed to remain constant and are taken at the datasheet reference temperature of 25°C.

B. Thermal Modelling

For the thermal analysis, the time-varying power loss profiles of the IGBT and diode are applied to a multi-stage Cauer-type RC thermal network representing the heat flow path from the device junctions to the ambient. Fig. 1 illustrates the simplified Cauer representation used to model this behaviour. In this network, the junction-to-case thermal dynamics of the IGBT and diode are described by the impedances Z_{IGBT} and Z_{Diode}, respectively, whereas the shared case to ambient path is represented by Z_{Paste} and Z_{Sink}, capturing heat transfer through the thermal interface material and the heat sink to the ambient temperature T_a. A multi-stage Cauer-type RC network is employed because it can approximate the complex thermal behaviour of the semiconductor module using multiple thermal nodes and RC elements that represent the distributed heat flow paths and thermal time constants within the device structure. At the same time, this representation remains computationally efficient, making it suitable for long-duration simulations while still capturing the dominant transient thermal dynamics of the IGBT and diode.

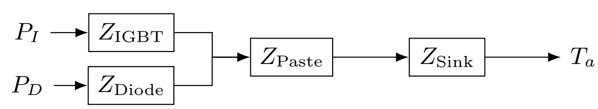

Fig. 1: Cauer-type thermal network of IGBT-diode module

The RC parameters used in the thermal model are derived from manufacturer provided transient thermal impedance data and therefore capture the dynamic thermal behaviour of the device across multiple thermal time constants. The semiconductor losses act as heat sources at the junction nodes of the IGBT and diode, and the generated heat propagates through the RC ladder via the package, thermal interface material and heat sink toward the ambient environment. The temperature evolution of each node is governed by the heat balance equation (1), where T_i and C_i denote the temperature and thermal capacitance of node i, $P_i(t)$ represents the heat generation at that node, and R_{ij} is the thermal resistance between node i and its neighbouring node j. This formulation results in a coupled system of ordinary differential equations describing the transient thermal behaviour of the network. Solving this system yields the junction temperature of the IGBT and diode under time-varying losses and ambient conditions, which serve as inputs for the subsequent lifetime assessment.

$$C_i \frac{dT_i}{dt} = P_i(t) + \sum_j \frac{T_j - T_i}{R_{ij}}, \tag{1}$$

C. Lifetime Estimation

The resulting junction temperature trajectories are post processed using rainflow counting to extract thermal cycles (amplitude, mean temperature and count). These cycles are mapped to cycles to failure using the power cycling lifetime model in (2) [12], where the model parameters capture technology scaling, thermo-mechanical fatigue, temperature activated ageing and time dependent degradation effects, as summarized in Table II. The lifetime is then estimated using Miner's linear damage rule, enabling lifetime estimation of the IGBT and diode under realistic operating conditions.

$$N_f = A_0 A_1^{e^{-\frac{\Delta T_j - T_0}{\lambda}}} (\Delta T)^{\alpha - e^{-\frac{\Delta T_j - T_0}{\lambda}}} \times e^{\frac{E_a}{k_B T_{jm}}} \left(\frac{C + t_{on}^{\gamma}}{C + 2^{\gamma}} \right) k_{\text{thickness}} \tag{2}$$

TABLE II: Parameters for power cycle model

Parameter	Value	Explanation
A_0	2.9×10^9	Technology Coefficient
A_1	60	Factor of Low ΔT Extension
T_0 [K]	40	Initial Temperature for Low ΔT Extension
λ [K]	17	Drop Constant of Low ΔT Extension
α	-4.3	Coffin–Manson Exponent
E_a [J]	4.50×10^{-20}	Activation Energy
k_B [J/K]	1.38×10^{-23}	Boltzmann Constant
C	1	Time Coefficient
γ	-0.75	Time Exponent
$k_{\text{thickness}}$	$1, .65, .5, .33$	Chip Thickness Factor

Further model explanations are provided in *Model_documentation_PEARL_IGBT_and_Diode.pdf* in the *IGBT and Diode* folder of the PEARL GitHub repository [13], covering the electrical loss, thermal and lifetime assessment models. A visual overview of the methodology is provided in Fig. 2, while a detailed description of the overall algorithm can be found in Algorithm 1. In practice, the IGBT-diode power module reliability assessment framework of PEARL requires only a small set of open and readily available inputs such as manufacturer datasheet parameters for the selected IGBT, diode, thermal interface material and heat sink, together with the inverter configuration and time-varying active and reactive power profiles.

III. Case Study

A. Case Study 1: Impact of Reactive power on module

Motivated by the growing requirement for power converters to provide reactive power support, this case study quantifies how reactive power provision by inverter-based resources affects the lifetime of IGBT-diode power modules. The simulation setup is illustrated in Fig. 3. In this scenario, the IBR operates at rated apparent power for a duration of 1 hour, with constant power factor values between 0 and 1 under both inductive and capacitive operating conditions. The mission profile is cyclically repeated over the service lifetime to evaluate the influence of reactive power on semiconductor lifetime. Although a simplified operating condition with constant rated power and constant ambient temperature is considered for demonstration purposes, the framework is fully capable of handling realistic, long-duration mission profiles with time-varying active power, reactive power and ambient temperature.

1. Mission profile & parameters: Operating scenario, semiconductor module and inverter configuration

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2. Power flow computation: Compute inverter operating conditions and instantaneous semiconductor currents

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3. Semiconductor power losses: Compute switching and conduction losses of IGBT and diode

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4. Multi-stage Cauer-type RC network: Dynamic IGBT and diode junction temperature computation

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5. Thermal Cycle Counting Extraction of junction temperature cycles using the rainflow algorithm

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6. Power cycling lifetime model: Estimate cycles to failure

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7. Lifetime estimation: Cumulative damage evaluation using Miner's rule

Fig. 2: Methodological workflow.

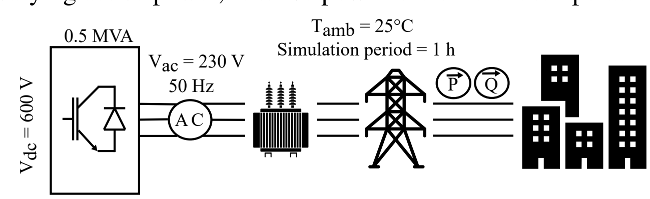

Fig. 3: Case study framework

The inverter is modeled as a three-phase converter rated at 0.5 MVA, with an AC side phase RMS voltage of 230 V and a DC link voltage of 600 V. It operates at a modulation index of 1 using space vector modulation with a switching frequency of 10 kHz and a grid frequency of 50 Hz. Each switch position is modeled as a power semiconductor module comprising ten identical IGBT with anti-parallel diode pairs connected in parallel to achieve the required current rating.

The semiconductor device considered is the 600 V, 50 A IKW50N60H3 IGBT with integrated anti-parallel diode from Infineon [14]. The junction-to-case transient thermal impedance characteristics of both the IGBT and diode are extracted and implemented as multi-stage Cauer-type RC networks to capture dynamic thermal behavior. In addition to thermal parameters, switching and conduction characteristics are obtained from the datasheet to parameterize the electrical loss model. The case-to-sink interface is modeled using Thermal Grizzly Kryonaut thermal paste with a thermal resistance of 0.0032 K/W [15]. The heat sink is represented by the aluminum Cooling Innovations 3-121211U device, with a specific heat of 900 J/kg K and thermal resistance of 2 K/W at air speed of 1 m/s [16]. All numerical parameter values used in this case study are listed in *Input_parameters.py* in the *IGBT and Diode* folder of the PEARL GitHub repository

Algorithm 1: Framework for lifetime estimation of IGBT-diode power modules

Input: Mission profile $P[k], Q[k], T_{\text{env}}[k]$; inverter and component parameters; thermal RC network (IGBT, diode, paste, sink); lifetime model constants; numerical controls.

Output: Electrical, thermal and lifetime data (Parquet); estimated lifetime for IGBT/diode; plots.

Initialize: Instantiate parameter object; derive electrical quantities $(S, V_s, I_s, \varphi, \text{pf}, V_{dc})$; create output directories; set thermal initial state $T_0 \leftarrow$ `None`.

Chunking: Divide the mission profile into segments of length `chunk_seconds`.

For each *chunk* c

- **Electrical loss evaluation:** Compute instantaneous device currents i_I and i_D, and evaluate conduction and switching losses P_I and P_D.
- **Thermal simulation (Cauer network):** Solve the coupled thermal ODE using a stiff solver (e.g., BDF) to obtain junction, case and sink temperatures.
- **Data storage:** Store electrical and thermal quantities as Parquet files.
- **Thermal cycle extraction:** Apply rainflow counting to $T_{j,I}(t)$ and $T_{j,D}(t)$ to obtain ΔT_j, mean temperature, cycle period and counts.
- **Lifetime evaluation:** Compute $N_{f,I}$ and $N_{f,D}$ using the lifetime model, and store lifetime results.
- **State update:** Update T_0 using the final thermal state of the current chunk.

Lifetime aggregation: Combine all chunks and evaluate accumulated damage using Miner's rule to obtain device lifetime.

Post-processing: Generate electrical, thermal and lifetime plots from stored data.

[13].

B. Case Study 2: Heat sink selection for power module

During inverter design, semiconductor devices must be paired with an adequately sized heat sink to ensure that the junction temperature does not exceed the manufacturer specified limit. In this study, the IKW50N60H3 IGBT-diode module [14] has a maximum allowable junction temperature of 175°C, which defines the thermal design constraint. To satisfy this requirement, heat sinks from the Cooling Innovations aluminum series 3-1212 [16] are evaluated. These heat sinks differ in height, weight and thermal resistance as a function of air speed, as summarized in Table III. For the thermal design assessment, all conditions are kept identical to Case Study 1 as described in Section III-A, with the only difference that pf is fixed at 1. This operating point is used to evaluate both the thermal feasibility and the corresponding lifetime impact of different heat sink configurations.

TABLE III: Heat sinks specifications

P/N	Area[mm^2]	Height[mm]	Weight[g]	Thermal Resistance [°C/W]			
				1m/s	2m/s	3m/s	4m/s
02U	967.21	5.1	7.0	11.6	7.34	5.39	4.27
04U	967.21	10.2	9.7	5.46	3.20	2.28	1.79
06U	967.21	15.2	12	3.62	2.08	1.48	1.16
08U	967.21	20.3	15	2.72	1.55	1.11	0.88
10U	967.21	25.4	18	2.20	1.25	0.90	0.72

IV. Results and Discussion

Fig. 4 represents the electro-thermal computation chain for a segment of the simulation using the mission profile, operating conditions, design configuration and component specifications defined in Case Study 1 in Section III-A. For clarity, initially only one fundamental grid cycle (20 ms at 50 Hz) is shown and two operating points ($pf = 1$ and $pf = 0$) are considered. The purpose of this illustration is to provide insight into the electro-thermal behaviour within a single grid cycle, including the instantaneous currents, the resulting total losses and the corresponding junction temperature response of the IGBT and diode.

Fig. 4(a) shows the instantaneous semiconductor currents reconstructed from the converter operating conditions. The current is shared between the IGBT and the anti-parallel diode according to the phase relation between voltage and current. When they are in phase ($pf = 1$), the IGBT conducts most of the current, whereas with increasing phase shift ($pf = 0$), the diode conduction becomes more pronounced.

Based on the IGBT-diode currents along with other parameters, the instantaneous power losses of the IGBT and diode are calculated, as shown in Fig. 4(b). The total semiconductor loss consists of conduction and switching components. Conduction losses arise from the current flowing through the device during its conduction interval and are determined by the device voltage and current magnitude, while switching losses occur during turn-on and turn-off transitions due to the overlap of voltage and current.

These time-varying power losses are applied to the multi-stage Cauer-type thermal network, which represents the transient junction temperatures of the IGBT and diode as shown in Fig. 4(c). At $pf = 1$, most of the current flows through the IGBT, resulting in higher IGBT losses and a noticeable temperature rise, while the diode temperature remains nearly constant due to its negligible losses. At $pf = 0$, the IGBT still dissipates more power overall, while the diode dissipates less but exhibits a higher temperature. This is due to the higher thermal resistance of the diode ($1.05\ \text{K/W}$) compared to the IGBT ($0.45\ \text{K/W}$) [14], meaning that for similar power dissipation the diode experiences a larger temperature rise. Additionally, the temperature peaks occur slightly after the power-loss peaks due to thermal inertia, and the temperature decreases gradually even after conduction stops because the device requires time to cool through the thermal path.

Fig. 5 shows the junction-temperature evolution of the IGBT and diode over the 1-hour simulation. As heat accumulates in the thermal network, the temperatures converge to a steady-state value, which remains constant under the fixed operating

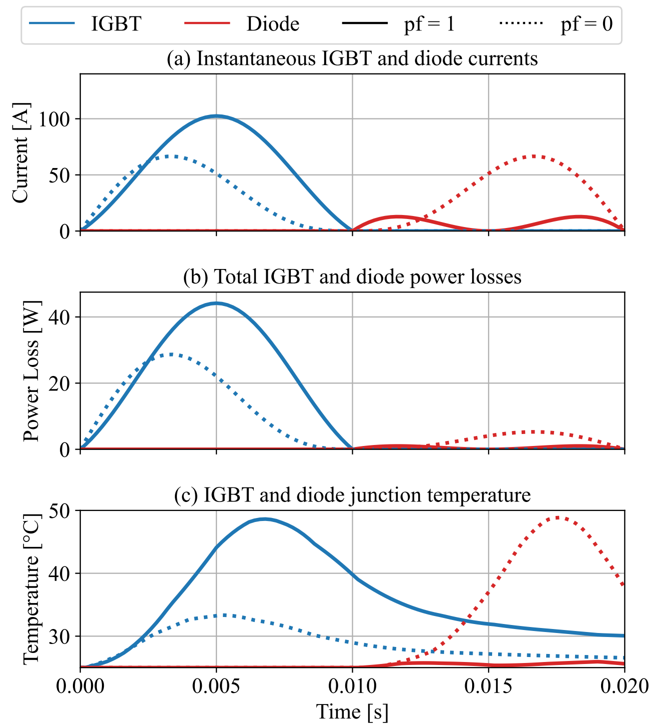

Fig. 4: Electro–thermal response of the IGBT-diode module: (a) Instantaneous device currents, (b) Total power losses, (c) Junction temperature

conditions of Case Study 1 and is used for the subsequent lifetime calculation.

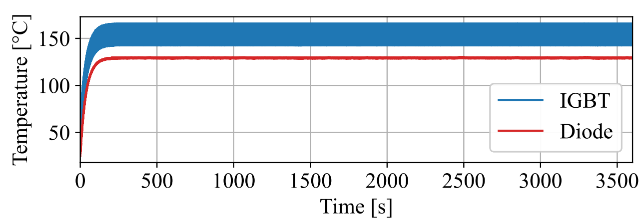

Fig. 5: Junction temperature evolution of IGBT and diode

A. Case Study 1 Results: Impact of Reactive power on module

Fig. 6 summarizes the influence of power factor on estimated lifetime of the IGBT-diode power module. The inverter is simulated at constant apparent power while the power factor is varied from 0 to 1 for both inductive and capacitive conditions, resulting in 20 operating scenarios. For each case, the average device current, total losses, junction temperature and estimated lifetime are extracted for comparison.

As shown in Fig. 6(a), the average device currents redistribute between the IGBT and its anti-parallel diode as the power factor varies, while the total phase current remains constant due to constant apparent power operation. At unity power factor, voltage and current are aligned and most of the current flows through the IGBT with limited diode conduction. As the phase angle increases toward 90° ($pf \rightarrow 0$), diode conduction intervals increase, leading to a complementary redistribution of current between the two devices.

The redistribution of average device currents between the IGBT and its anti-parallel diode with varying power factor directly influences the semiconductor losses, as shown in Fig. 6(b). Near unity power factor, the IGBT conducts for longer intervals, leading to higher average IGBT losses and reduced diode dissipation, whereas as the power factor approaches zero, a larger fraction of the current returns through the diode, increasing its conduction losses and reducing those of the IGBT. The loss trends are not perfectly symmetric because the two devices exhibit different electrical characteristics. The IGBT incurs both conduction and switching losses, including turn-on and turn-off energy components that scale nonlinearly with current, while the diode losses are primarily governed by conduction and reverse recovery effects. Consequently, the total power dissipation, defined as the sum of switching and conduction losses, evolves differently for the two devices as the power factor varies. Further details on the underlying loss modeling and device physics are provided in *Model_documentation_PEARL_IGBT_and_Diode.pdf* [13].

The resulting thermal response due to the device power losses is shown in Fig. 6(c). The junction temperature rise is approximately proportional to the dissipated power, with the proportionality factor given by the thermal resistance. For the considered module, the thermal resistance is 0.45 K/W for the IGBT and 1.05 K/W for its antiparallel diode [14]. Hence, each watt of dissipated power results in about 0.45 K and 1.05 K temperature rise for the IGBT and diode, respectively. However, the IGBT and diode are thermally coupled through the thermal paste and heat sink (see Fig. 1). Therefore, the junction temperature of each device is influenced not only by its own losses but also by the losses of the other device. At pf $= 1$, the IGBT exhibits higher power losses, leading to a higher junction temperature. Although the diode losses are lower in this condition, its larger thermal resistance results in a comparatively stronger temperature rise per watt, and it is additionally heated by the elevated case temperature caused by the IGBT losses. As the power factor approaches zero, the IGBT losses decrease while the diode losses increase. Nevertheless, the strong reduction in IGBT heating lowers the common case temperature, which partially offsets the diode's increasing self-heating. At $pf = 0$, where both devices dissipate comparable power, the diode reaches a higher junction temperature due to its significantly larger thermal resistance.

Finally, the estimated lifetimes of the IGBT and diode are illustrated in Fig. 6(d). Since semiconductor lifetime is strongly governed by junction temperature, the different thermal responses of the IGBT and diode lead to opposite lifetime trends. It should be noted that the effect of temperature on the lifetime of the IGBT and diode differs due to their slightly distinct lifetime model parameters and the differences in temperature magnitude. At power factors close to unity, higher IGBT temperatures lead to reduced IGBT lifetime, while the diode experiences comparatively lower thermal stress and therefore a longer lifetime. As the power factor approaches zero, longer diode conduction intervals increase diode temperature and reduce its lifetime, while the corresponding reduction in IGBT thermal stress improves the IGBT lifetime. These results demonstrate that, even under constant apparent power operation, reactive power provision significantly alters device-level thermal stress and lifetime consumption.

Fig. 6(d) presents the individual lifetimes of the IGBT and diode. However, since failure of either device leads to failure of the entire module, the effective lifetime is determined by the minimum of the two. Accordingly, Fig. 7 shows the module lifetime as a function of power factor. It can be observed that at both $pf = 0$ and $pf = 1$, the module lifetime is approximately one year, limited by the IGBT at $pf = 1$ due to high current stress under rated active power, and by the diode at $pf = 0$ due to increased conduction and thermal stress under purely reactive operation. While reduced lifetime at rated power operation at $pf = 1$ is expected, notably even at $pf = 0$ the module experiences significant degradation despite zero active power transfer, primarily due to high reverse current stress in the diode. This behaviour aligns with prior studies indicating that reactive power operation accelerates wear out in power electronic converters [17], [18].

As Fig. 7 illustrates the maximum module lifetime is achieved at approximately $pf = 0.4$, where a balanced distribution of electrical and thermal stress between the IGBT and diode results in moderate junction temperature swings and mean temperatures. In this case study, the Infineon IKW50N60H3 IGBT module with an integrated anti-parallel diode [14] was used, however, the results are device-specific and different semiconductor modules may exhibit different lifetime characteristics due to different component specifications under similar conditions. This demonstrates the capability of the proposed framework to evaluate and compare semiconductor modules prior to implementation, enabling more informed and reliability-oriented design decisions.

It can be observed from Fig. 6 that inductive and capacitive operation at the same power factor produce nearly identical results. The small deviations visible in Fig. 6(c) and Fig. 6(d) arise from the numerical solution of the Cauer thermal model using ordinary differential equations and solver tolerances. Although higher numerical accuracy could reduce these deviations, it would significantly increase the computational effort therefore, a balanced setting between accuracy and computation time was selected.

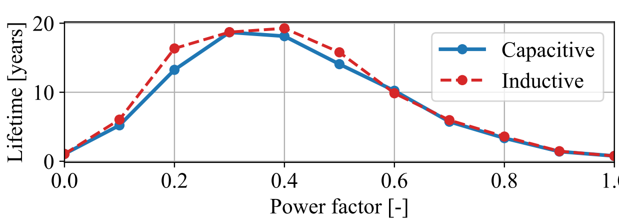

Fig. 7: Estimated lifetime of the semiconductor power module

B. Case Study 2 Results: Heat sink selection for power module

Fig. 8 shows the maximum junction temperature of the IGBT for different heat sink configurations. The IGBT is specifically considered, as it is evident from Fig. 6(c) that at

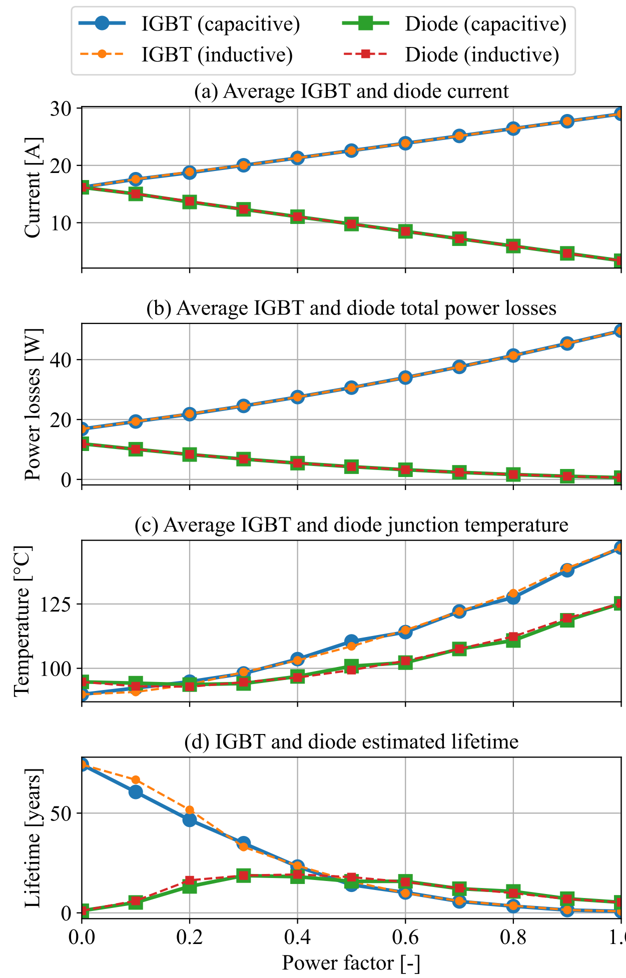

Fig. 6: Influence of power factor on the electro-thermal behavior and lifetime of the IGBT-diode power module: (a) average device current, (b) average total semiconductor power losses, (c) average junction temperature and (d) estimated lifetime

$pf = 1$ it exhibits higher junction temperatures than the diode, making it the limiting component. The heat sinks are evaluated for air velocities between 1 m/s and 4 m/s, which affect the heat sink thermal resistance. Smaller heat sinks (02U, 04U) have higher thermal resistance due to their limited surface area (see Table III), resulting in higher junction temperatures that can exceed the allowable limit of 175°C. Increasing the heat sink size (06U, 08U, 10U) reduces thermal resistance, improves heat dissipation and keeps the junction temperature within the safe operating range.

Furthermore, Fig. 8 illustrates the trade off between heat sink size and the required air flow for thermally safe operation. Smaller heat sinks require higher air velocities to keep the junction temperature below the allowable limit. For example, the 04U heat sink operates safely only at around 4 m/s, but its compact height (10.2 mm) results in a small volume of about $9865.54\ \mathrm{mm}^3$, roughly 2.5 times smaller than the larger 10U heat sink ($24567.13\ \mathrm{mm}^3$), which can maintain safe temperatures at air speeds slightly above 1 m/s. This highlights the design trade-off that compact heat sinks save space but require stronger forced cooling implying larger fans and increased auxiliary power consumption, while larger heat sinks reduce airflow requirements and auxiliary power consumption.

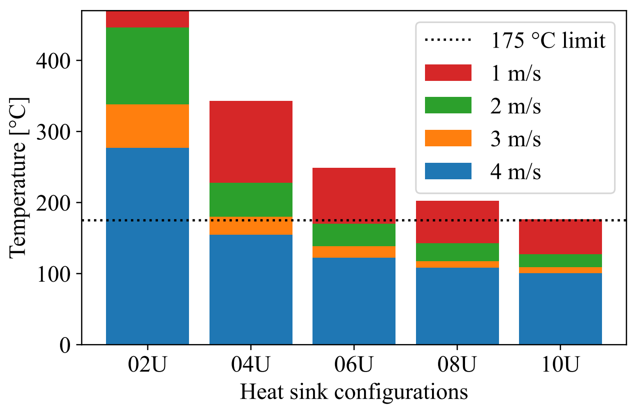

Fig. 8: Junction temperature for different heat sink configurations

Fig. 8 also identifies the heat sink configurations that maintain the IGBT-diode power module below the required temperature limit of 175°C and provides the corresponding IGBT junction temperature. Based on this, Fig. 9 presents the resulting module lifetime for different heat sink configurations, indicating a clear trend that lower junction temperatures lead to longer lifetimes. Among the evaluated configurations, the 10U heat sink at 4 m/s results in the lowest junction temperature as shown in Fig. 8 and correspondingly achieves the highest lifetime as illustrated in Fig. 9. Heat sink configurations not shown in Fig. 9 correspond to cases where the junction temperature exceeds 175°C as evident from Fig. 8, leading to immediate failure and hence zero lifetime. Overall, the results clearly demonstrate that higher junction temperatures accelerate lifetime degradation of the semiconductor module, while lower temperatures significantly enhance its lifetime, consistent with established power-electronics lifetime behavior.

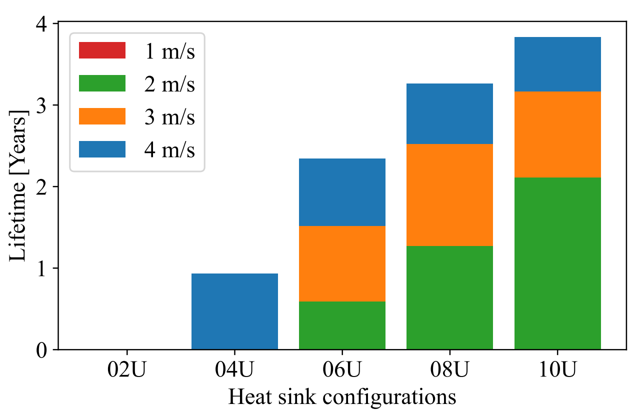

Fig. 9: Estimated lifetime for different heat sink configurations

These results highlight the strong coupling between thermal design and semiconductor lifetime, showing that even under identical electrical operating conditions, inadequate heat-sink performance can lead to elevated junction temperatures and accelerated lifetime degradation. The presented analysis therefore demonstrates how the PEARL framework can support inverter design decisions by quantitatively evaluating the thermal feasibility and reliability implications of different heat sink configurations. Although practical inverter design involves many additional considerations, IGBT-diode power module reliability assessment framework of PEARL provides a physics-based assessment of semiconductor thermal behavior and helps identify which combinations of heat sink size, cooling conditions and operating points are thermally viable, and how their corresponding impact on semiconductor module lifetime.

V. Conclusion

This paper presents an IGBT-diode power module reliability assessment framework of PEARL, which integrates electrical loss modeling, dynamic junction temperature modeling and power cycling lifetime estimation within a unified workflow for IGBT-diode power modules. Using parameters derived directly from manufacturer datasheets, the framework translates mission profile operating conditions into cumulative lifetime consumption, enabling systematic evaluation of how operating conditions affect semiconductor reliability.

Two case studies were conducted to demonstrate the capabilities of the framework. The first case study examined the impact of reactive power operation on semiconductor lifetime and showed that, even at constant apparent power, variations in power factor can lead to different lifetimes for the IGBT and diode. The second case study evaluated the influence of heat sink size and air flow conditions on the maximum junction temperature of the semiconductor module, and demonstrated how these thermal design choices directly impact the resulting semiconductor lifetime, thereby supporting reliability-aware design decisions in the early stages of inverter development.

Overall, the results demonstrate that the IGBT-diode power module reliability assessment framework developed within PEARL provides a transparent framework for linking time-varying mission profiles, operating conditions, design configurations and component specifications to semiconductor lifetime estimation, enabling early stage evaluation of reliability implications for different design choices.

VI. Future Outlook

Future developments will extend the framework to additional power electronic components such as DC-link capacitors, gate driver circuits, LCL filters, MOSFETs, etc. By incorporating lifetime models for these components, PEARL can be expanded toward system-level reliability analysis of grid connected converters. In the long-term, the framework aims to evolve into a flexible reliability assessment platform capable of modeling diverse power electronic based assets such as STATCOMs, HVDC converters and other converter-dominated infrastructure, enabling evaluation of grid wide reliability under varying operating conditions.

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